Project 5

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Contents

Assuming the single index model

$$R_i = \alpha_i + \beta_i R_m + \epsilon_i, \qquad \mathrm{cov}(\epsilon_i, R_m) = 0, \qquad \mathrm{cov}(\epsilon_i, \epsilon_j) = 0 \ \ (i \neq j)$$

on the same 30 stocks with `^GSPC` as R_m. Two periods are needed here:

```
#> period 1  01-Jan-2017 to 01-Jan-2022   59 returns
```

```
#> period 2  01-Jan-2022 to 31-Jul-2026   54 returns
```

Period 1 is the training file, so the estimates there are Project 4's.

(a)

The model gives Σ from the betas and residual variances alone,

$$\sigma_{ij} = \beta_i \beta_j \sigma_m^2 \ \ (i \neq j), \qquad \sigma_{ii} = \beta_i^2 \sigma_m^2 + \sigma_{\epsilon_i}^2$$

and with short sales allowed the optimum is

$$\mathbf{Z} = \Sigma^{-1}(\bar{R} - R_f \mathbf{1}), \qquad x_i = \frac{z_i}{\sum_j z_j}$$

```
#> stocks with positive beta: 30 of 30
```

```
#> beta range 0.363 (LLY) to 1.978 (TSLA)
```

```
#> Rf = 0.0010 per month
```

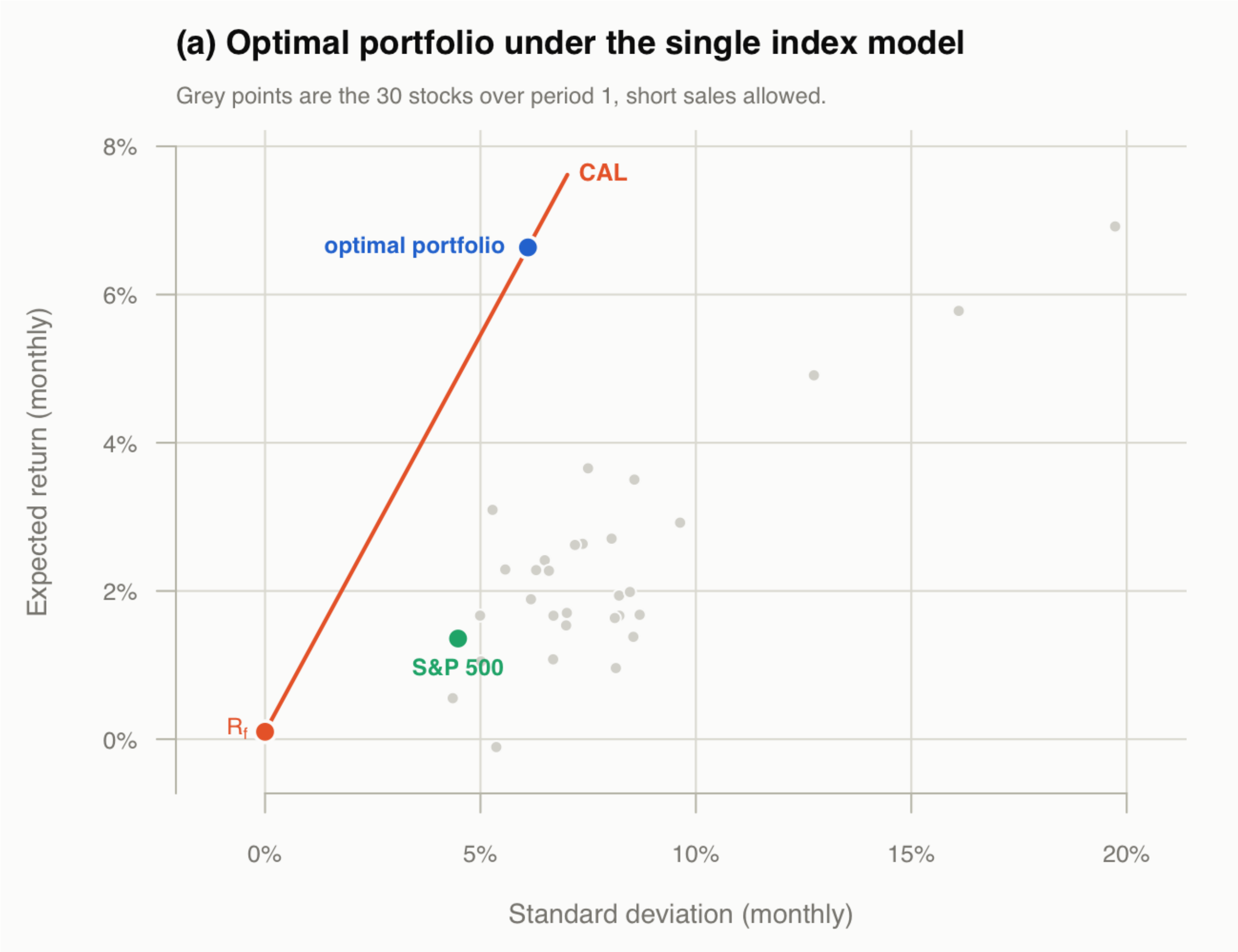
Every beta is positive, so the filter removes nothing.

Table 1: Composition of the optimal portfolio

Stock	Weight	Beta
NVDA	0.1110	1.3118
AAPL	0.1657	1.2019
MSFT	0.6475	0.8625
AVGO	0.1359	0.9597

Stock	Weight	Beta
AMD	0.0720	1.8741
NOW	0.2819	1.0001
GOOG	0.1361	1.0576
META	-0.0631	1.2828
NFLX	0.0993	0.8136
VZ	-0.1170	0.4422
DIS	-0.1768	1.1871
T	-0.4615	0.7345
V	0.0308	0.9692
JPM	-0.1304	1.1351
MS	-0.1808	1.5381
GS	-0.2637	1.4983
BAC	-0.2400	1.4827
SCHW	-0.0354	1.0448
AMZN	0.0971	1.0959
TSLA	0.0610	1.9779
MCD	0.1671	0.5982
SBUX	0.0245	0.8235
HD	0.1319	0.9825
EBAY	-0.0399	1.1172
LLY	0.2156	0.3633
NVO	0.2937	0.4391
UNH	0.1330	0.8612
JNJ	-0.1312	0.7387
AMGN	-0.0203	0.6394
PFE	0.0560	0.6977

```
#> C*             = 0.017356
#> E              = 0.066358  (6.636% per month)
#> sd             = 0.061025  (6.103% per month)
#> Sharpe         = 1.0710
#> portfolio beta = 0.4927
#> short positions = 12 of 30, sum|x| = 4.72
```



The same vector two other ways, as checks:

```
#> max |Sherman-Morrison - solve()|  5.33e-15
#> max |cut-off form - Z|            5.33e-15
```

Σ is diagonal plus rank one, so Sherman–Morrison inverts it in closed form. Expanding that inverse gives the cut-off form of handout #13,

$$z_i = \frac{\beta_i}{\sigma^2_{\epsilon_i}} \left(\frac{\bar{R}_i - R_f}{\beta_i} - C^* \right), \qquad C^* = \frac{\sigma^2_m \sum_j \frac{(\bar{R}_j - R_f)\beta_j}{\sigma^2_{\epsilon_j}}}{1 + \sigma^2_m \sum_j \frac{\beta^2_j}{\sigma^2_{\epsilon_j}}}$$

With short sales allowed there is no cut-off to apply: every stock enters, long if $(\bar{R}_i - R_f)/\beta_i$ exceeds C^* and short if not.

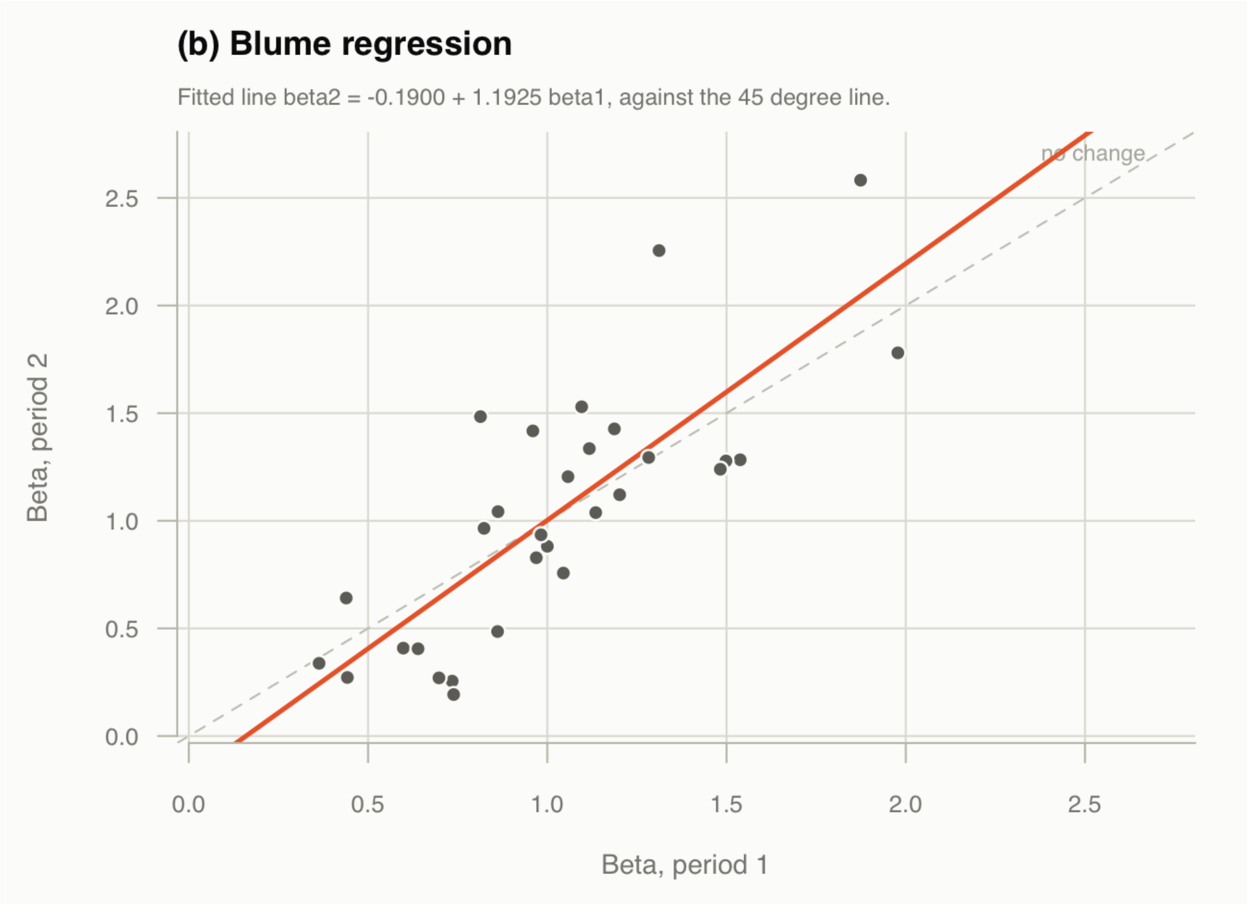
(b) Blume

$$\beta_{i,2} = a + b\,\beta_{i,1} \quad \longrightarrow \quad \hat{\beta}_{i,3} = a + b\,\beta_{i,2}$$

```
#> beta2 = -0.1900 + 1.1925 beta1   (R^2 = 0.636)

#>
#>              mean      sd
```

```
#> period 1   1.0244  0.3910
#> period 2   1.0315  0.5847
#> forecast   1.0401  0.6973
```



(b) Vasicek

$$\beta_i^{adj} = \frac{\sigma^2_{\beta_i}}{\sigma^2_{\beta_i} + \sigma^2_{\beta_1}}\,\bar{\beta}_1 \;+\; \frac{\sigma^2_{\beta_1}}{\sigma^2_{\beta_i} + \sigma^2_{\beta_1}}\,\beta_{i,1}$$

where $\sigma^2_{\beta_i}$ is the squared standard error of stock i's beta in period 1 and $\sigma^2_{\beta_1}$ the variance of the betas across stocks.

```
#> beta_bar                 = 1.0244
#> cross-sectional variance = 0.15284
#> weight on the mean: 0.069 to 0.640 (median 0.158)
#> beta sd: raw 0.3910 -> adjusted 0.2813
```

Table 2: Raw betas by period, the Vasicek adjustment of period 1, and the Blume forecast

Stock	Beta p1	Beta p2	Vasicek	Blume forecast
NVDA	1.3118	2.2554	1.1904	2.4996

Stock	Beta p1	Beta p2	Vasicek	Blume forecast
AAPL	1.2019	1.1209	1.1659	1.1467
MSFT	0.8625	1.0425	0.8736	1.0532
AVGO	0.9597	1.4176	0.9707	1.5005
AMD	1.8741	2.5822	1.4330	2.8894
NOW	1.0001	0.8815	1.0043	0.8612
GOOG	1.0576	1.2049	1.0543	1.2468
META	1.2828	1.2939	1.2403	1.3530
NFLX	0.8136	1.4839	0.8795	1.5796
VZ	0.4422	0.2720	0.4884	0.1344
DIS	1.1871	1.4269	1.1580	1.5116
T	0.7345	0.2550	0.7615	0.1141
V	0.9692	0.8282	0.9747	0.7977
JPM	1.1351	1.0380	1.1222	1.0478
MS	1.5381	1.2834	1.4755	1.3405
GS	1.4983	1.2779	1.4329	1.3339
BAC	1.4827	1.2395	1.4284	1.2881
SCHW	1.0448	0.7570	1.0407	0.7127
AMZN	1.0959	1.5295	1.0824	1.6340
TSLA	1.9779	1.7801	1.3677	1.9329
MCD	0.5982	0.4083	0.6375	0.2969
SBUX	0.8235	0.9648	0.8539	0.9606
HD	0.9825	0.9349	0.9868	0.9249
EBAY	1.1172	1.3353	1.0964	1.4024
LLY	0.3633	0.3378	0.5082	0.2128
NVO	0.4391	0.6410	0.5179	0.5744
UNH	0.8612	0.4855	0.8841	0.3889
JNJ	0.7387	0.1927	0.7602	0.0398
AMGN	0.6394	0.4055	0.7058	0.2935
PFE	0.6977	0.2702	0.7576	0.1321

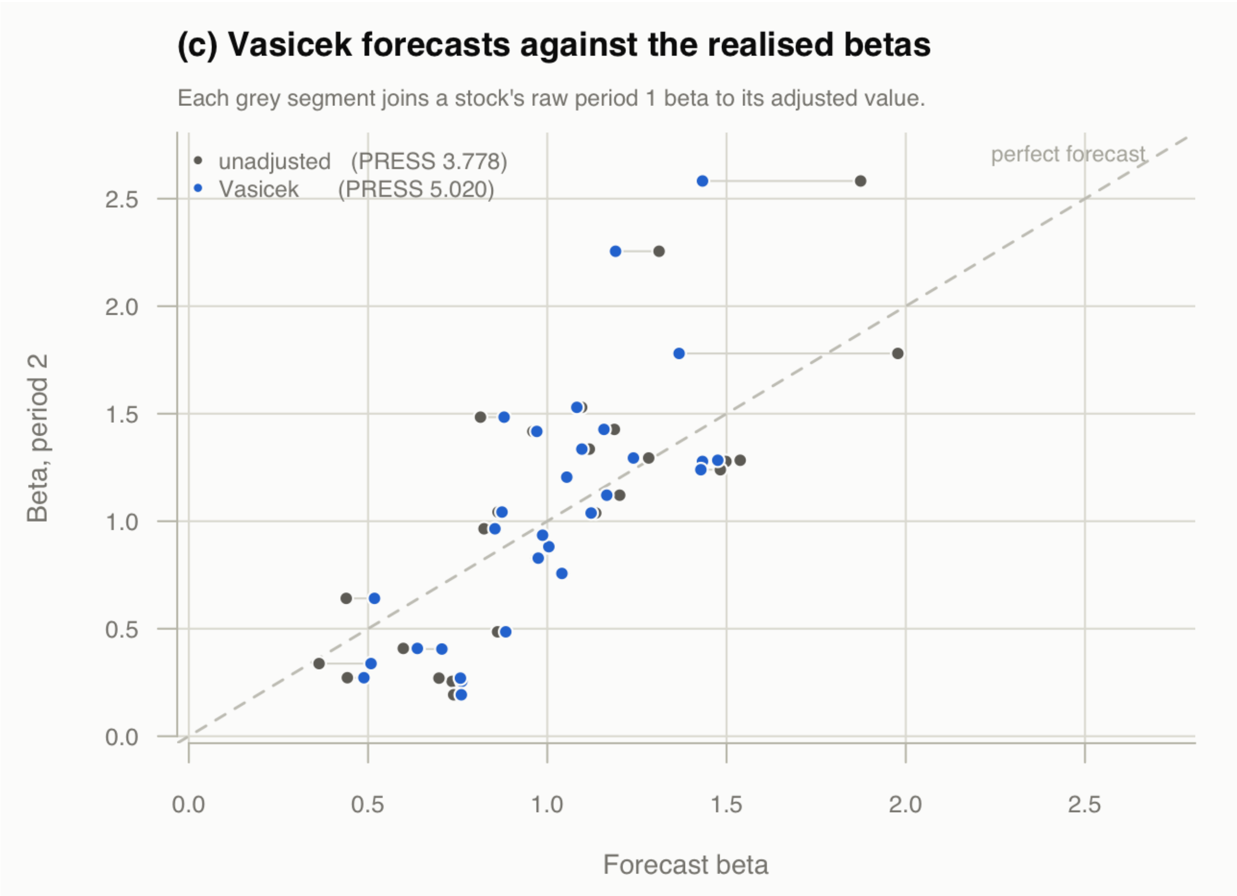
(c)

$$\text{PRESS} = \sum_{i=1}^{n} \left(\beta_i^{adj} - \beta_{i,2} \right)^2$$

taking the sum of squared forecast errors, as in Homework 4 exercise 1.

```
#> Vasicek     5.0201   (mean square 0.167337)
```

```
#> unadjusted  3.7782   (mean square 0.125941)
```



The unadjusted period 1 betas are shown alongside as a reference point; their PRESS of 3.7782 is the one decomposed in Homework 4 exercise 1. The betas in this sample dispersed rather than reverting — their cross-sectional standard deviation rose from 0.391 to 0.585 — which is also why the Blume slope came out above one.

Appendix: full code

```
# Project 5, parts a-c: optimisation under the single index model, and the
# Blume and Vasicek beta adjustments.

local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "beta_adjust.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."

# ---- data ------------------------------------------------------------------
# Period 1 is the training file, the same 59 returns Projects 1, 2 and 4 use, so
# the betas here are Project 4's betas. Period 2 is the test file.

read_ret <- function(f) {
  a <- read.csv(file.path(DATA, f), sep = ",", header = TRUE, check.names = FALSE)
  stopifnot(!any(duplicated(names(a))))
```

```r
  P <- as.matrix(a[, -1][, -1])
  stopifnot(ncol(P) == 31, !anyNA(P))
  n <- nrow(P)
  P[-1, ] / P[-n, ] - 1
}

ret1 <- read_ret("stockData_train.csv")     # 01-Jan-2017 to 01-Jan-2022
ret2 <- read_ret("stockData_test.csv")      # 01-Jan-2022 to 31-Jul-2026
stopifnot(nrow(ret1) == 59, nrow(ret2) == 54)

stk <- setdiff(colnames(ret1), "^GSPC")
stopifnot(length(stk) == 30, identical(stk, setdiff(colnames(ret2), "^GSPC")))

# ---- single index fits in each period ---------------------------------------

fit_period <- function(r) {
  Rm  <- as.numeric(r[, "^GSPC"])
  r30 <- r[, stk]
  fit <- lapply(stk, function(s) lm(r30[, s] ~ Rm))
  names(fit) <- stk
  list(
    Rm    = Rm,
    Rbar  = colMeans(r30),
    alpha = vapply(fit, function(f) unname(coef(f)[1]), numeric(1)),
    beta  = vapply(fit, function(f) unname(coef(f)[2]), numeric(1)),
    se    = vapply(fit, function(f) unname(summary(f)$coefficients[2, 2]), numeric(1)),
    sig2e = vapply(fit, function(f) summary(f)$sigma^2, numeric(1)),
    sd    = apply(r30, 2, sd),
    Tn    = nrow(r30)
  )
}

p1 <- fit_period(ret1)
p2 <- fit_period(ret2)

stopifnot(max(abs(p1$beta - apply(ret1[, stk], 2,
                  function(x) cov(x, p1$Rm) / var(p1$Rm)))) < 1e-12)
stopifnot(max(abs(p1$alpha - (p1$Rbar - p1$beta * mean(p1$Rm)))) < 1e-15)

# ---- (a) optimal portfolio, Z = Sigma^-1 R ----------------------------------

Rf  <- 0.001
pos <- p1$beta > 0

cat("(a) stocks with positive beta:", sum(pos), "of", length(pos), "\n")

sel   <- stk[pos]
beta  <- p1$beta[pos]
sig2e <- p1$sig2e[pos]
Rbar  <- p1$Rbar[pos]
sig2m <- var(p1$Rm)

Sigma <- outer(beta, beta) * sig2m
```

```r
diag(Sigma) <- beta^2 * sig2m + sig2e
stopifnot(isSymmetric(Sigma), min(eigen(Sigma, only.values = TRUE)$values) > 0)

Rex <- Rbar - Rf
Z   <- solve(Sigma, Rex)
x   <- Z / sum(Z)

# Sherman-Morrison: Sigma is diagonal plus rank one, so its inverse is explicit
D_inv <- diag(1 / sig2e)
sm    <- D_inv - sig2m * (D_inv %*% outer(beta, beta) %*% D_inv) /
                as.numeric(1 + sig2m * t(beta) %*% D_inv %*% beta)
stopifnot(max(abs(sm %*% Rex - Z)) < 1e-9)

# The cut-off form of the same solution, handout #13
C_star <- sig2m * sum(Rex * beta / sig2e) / (1 + sig2m * sum(beta^2 / sig2e))
z_egp  <- (beta / sig2e) * (Rex / beta - C_star)
stopifnot(max(abs(z_egp - Z)) < 1e-9)

E_G      <- sum(x * Rbar)
sd_G     <- sqrt(as.numeric(t(x) %*% Sigma %*% x))
sharpe_G <- (E_G - Rf) / sd_G
beta_G   <- sum(x * beta)

# No reweighting beats it
set.seed(183)
stopifnot(sharpe_G >= max(replicate(4000, {
  w <- x + rnorm(length(x), 0, 0.02); w <- w / sum(w)
  (sum(w * Rbar) - Rf) / sqrt(as.numeric(t(w) %*% Sigma %*% w))
})))

cat(sprintf("    Rf = %.4f, C* = %.6f\n", Rf, C_star))
cat(sprintf("    E %.4f%%, sd %.4f%%, Sharpe %.4f, portfolio beta %.4f\n",
            E_G * 100, sd_G * 100, sharpe_G, beta_G))
cat(sprintf("    %d short positions, sum|x| = %.2f\n\n", sum(x < 0), sum(abs(x))))

# ---- (b) Blume ---------------------------------------------------------------
#   beta2 = a + b beta1, then push the period 2 betas through the fitted line

blume     <- lm(p2$beta ~ p1$beta)
b_int     <- unname(coef(blume)[1])
b_slope   <- unname(coef(blume)[2])
b_r2      <- summary(blume)$r.squared
beta_next <- b_int + b_slope * p2$beta

# OLS passes through the means
stopifnot(abs((b_int + b_slope * mean(p1$beta)) - mean(p2$beta)) < 1e-12)

cat(sprintf("(b) Blume: beta2 = %.4f + %.4f beta1   (R^2 = %.3f)\n",
            b_int, b_slope, b_r2))
cat(sprintf("    beta mean/sd:  p1 %.4f/%.4f   p2 %.4f/%.4f   forecast %.4f/%.4f\n\n",
            mean(p1$beta), sd(p1$beta), mean(p2$beta), sd(p2$beta),
            mean(beta_next), sd(beta_next)))
```

```r
# ---- (b) Vasicek ------------------------------------------------------------
#   beta_adj = (var_bi/(var_bi+var_b1)) beta_bar + (var_b1/(var_bi+var_b1)) beta_i

var_bi   <- p1$se^2                          # squared standard error of each beta
var_b1   <- var(p1$beta)                     # cross-sectional variance of the betas
beta_bar <- mean(p1$beta)
w_prior  <- var_bi / (var_bi + var_b1)
beta_vas <- w_prior * beta_bar + (1 - w_prior) * p1$beta

stopifnot(all(w_prior > 0 & w_prior < 1))
stopifnot(all(abs(beta_vas - beta_bar) <= abs(p1$beta - beta_bar) + 1e-12))

cat("(b) Vasicek\n")
cat(sprintf("    beta_bar %.4f, cross-sectional variance %.5f\n", beta_bar, var_b1))
cat(sprintf("    weight on the mean: %.3f to %.3f (median %.3f)\n",
            min(w_prior), max(w_prior), median(w_prior)))
cat(sprintf("    beta sd: raw %.4f -> adjusted %.4f\n\n", sd(p1$beta), sd(beta_vas)))

# ---- (c) PRESS --------------------------------------------------------------

# Sum of squared forecast errors, the Klemkosky and Martin (1975) form used in
# Homework 4 exercise 1.
press_vas <- sum((beta_vas - p2$beta)^2)
press_raw <- sum((p1$beta  - p2$beta)^2)

# The unadjusted figure is the PRESS Homework 4 computes, so it must agree
stopifnot(abs(press_raw - 3.778242) < 1e-5)

cat("(c) PRESS against the realised period 2 betas\n")
cat(sprintf("    Vasicek       %.4f   (mean square %.6f)\n",
            press_vas, press_vas / length(stk)))
cat(sprintf("    unadjusted    %.4f   (mean square %.6f)\n\n",
            press_raw, press_raw / length(stk)))

betas <- data.frame(stock = stk, beta1 = p1$beta, beta2 = p2$beta,
                    vasicek = beta_vas, blume_next = beta_next, row.names = NULL)

# ---- plotting helpers -------------------------------------------------------

surface <- "#fcfcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_front <- "#2a78d6"; c_eq <- "#eb6834"; c_spx <- "#1baf7a"; c_arb <- "#6f6d67"

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}
ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
       cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}
pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
```

```r
}

# ---- (a) the optimal portfolio and the capital allocation line --------------

open_png("fig_tangency.png", h = 5.0)
xr <- c(-0.012, max(p1$sd) * 1.04)
yr <- c(-0.004, max(c(p1$Rbar, E_G)) * 1.14)
xtick <- pretty(c(0, max(p1$sd)))
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = xtick, col = grid, lwd = 1)

cal_x <- sd_G * 1.15
segments(0, Rf, cal_x, Rf + sharpe_G * cal_x, col = c_eq, lwd = 2)
text(cal_x, Rf + sharpe_G * cal_x, "CAL", pos = 4, offset = 0.3, cex = 0.75,
     font = 2, col = c_eq)

points(p1$sd, p1$Rbar, pch = 21, bg = "#d9d8d2", col = surface, lwd = 1, cex = 0.85)
pt(sd(p1$Rm), mean(p1$Rm), c_spx, "S&P 500", 1)
pt(0, Rf, c_eq, expression(R[f]), 2, off = 0.45)
pt(sd_G, E_G, c_front, "optimal portfolio", 2)

ax(1, xtick, sprintf("%.0f%%", xtick * 100))
ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(a) Optimal portfolio under the single index model", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.05, font = 2)
mtext("Grey points are the 30 stocks over period 1, short sales allowed.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

# ---- (b) Blume ---------------------------------------------------------------

open_png("fig_blume.png")
rr <- range(c(p1$beta, p2$beta)) + c(-0.12, 0.12)
plot(NA, xlim = rr, ylim = rr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(rr), v = pretty(rr), col = grid, lwd = 1)
abline(0, 1, col = "#c9c8c0", lwd = 1, lty = 2)
text(rr[2], rr[2], "no change", pos = 2, offset = 0.3, cex = 0.7, col = "#b4b3ab")
abline(b_int, b_slope, col = c_eq, lwd = 2.4)
points(p1$beta, p2$beta, pch = 21, bg = c_arb, col = surface, lwd = 1.2, cex = 1.05)

ax(1, pretty(rr), sprintf("%.1f", pretty(rr)))
ax(2, pretty(rr), sprintf("%.1f", pretty(rr)))
mtext("Beta, period 1", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Beta, period 2", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(b) Blume regression", 3, line = 1.8, adj = 0, col = ink, cex = 1.05, font = 2)
mtext(sprintf("Fitted line beta2 = %.4f + %.4f beta1, against the 45 degree line.",
              b_int, b_slope), 3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

# ---- (c) Vasicek forecasts against the realised betas ----------------------
```

```r
open_png("fig_vasicek.png")
rr <- range(c(p1$beta, p2$beta, beta_vas)) + c(-0.12, 0.12)
plot(NA, xlim = rr, ylim = rr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(rr), v = pretty(rr), col = grid, lwd = 1)
abline(0, 1, col = "#c9c8c0", lwd = 1.4, lty = 2)
text(rr[2], rr[2], "perfect forecast", pos = 2, offset = 0.3, cex = 0.7, col = "#b4b3ab")
segments(p1$beta, p2$beta, beta_vas, p2$beta, col = "#dcdbd4", lwd = 1)
points(p1$beta, p2$beta, pch = 21, bg = c_arb, col = surface, lwd = 1.1, cex = 0.95)
points(beta_vas, p2$beta, pch = 21, bg = c_front, col = surface, lwd = 1.1, cex = 0.95)

legend("topleft", bty = "n", cex = 0.72, text.col = muted,
       pch = 21, pt.bg = c(c_arb, c_front), col = surface, pt.lwd = 1.1,
       legend = c(sprintf("unadjusted   (PRESS %.3f)", press_raw),
                  sprintf("Vasicek      (PRESS %.3f)", press_vas)))

ax(1, pretty(rr), sprintf("%.1f", pretty(rr)))
ax(2, pretty(rr), sprintf("%.1f", pretty(rr)))
mtext("Forecast beta", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Beta, period 2", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(c) Vasicek forecasts against the realised betas", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.05, font = 2)
mtext("Each grey segment joins a stock's raw period 1 beta to its adjusted value.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

cat("wrote fig_tangency.png, fig_blume.png, fig_vasicek.png\n")
```